

*Class notes Problem*  $RE, RL, RL$

# CS & IT ENGINEERING



## Theory of Computation

### REGULAR LANGUAGE

**DPP – 01 Discussion Notes**

**By– Rohit Chauhan sir**



# [MCQ]



#Q. Which of the following language is non-regular?

**A**

$$L = \{wxw^R \mid x, w \in \{a, b\}^*\}.$$

$$w = ab \quad w^R = ba$$
$$w = \epsilon \quad w^R = \epsilon$$

$$\epsilon \cdot (\underbrace{ab}_{w} \overset{x}{ab}) \cdot \epsilon$$

$$\Rightarrow \epsilon \cdot (x) w^R(\epsilon)$$
$$\Rightarrow \underline{(a+b)^*}$$

**B**

$$L = \{wxw \mid w, x \in \{a, b\}^*\}.$$

$$wxw$$

$$\begin{cases} w = \epsilon \\ w = \epsilon \end{cases}$$

$$\epsilon \cdot x \cdot \epsilon = x = \underline{(a+b)^*}$$

**C**

$$L = \{wxwx \mid w, x \in \{a, b\}^+\}.$$

$$\textcircled{\textcircled{X} \textcircled{X}}$$

$$\underline{ab \ ab}$$

**D**

None of these.

C

# [MCQ]



#Q. Consider the following grammar  $G_1$  and  $G_2$ :

$G_1$ :  $S \rightarrow aAb$

$A \rightarrow aB \mid \epsilon$

$B \rightarrow Ab$

$G_2$ :  $S \rightarrow aABb$

$A \rightarrow aA \mid \epsilon$

$B \rightarrow bB \mid \epsilon$

$aAb = ab$

If  $G$  is RL then it will generate RL  
" " " Not RL it may/may not generate RL

$aAb \Rightarrow aabbb$   
 $\bar{a}aA\bar{b}b$   
 $\bar{a}a\bar{a}B\bar{b}b$   
 $\bar{a}a\bar{a}A\bar{b}b\bar{b}b \Rightarrow \#a's = \#b's$   
 $a^n b^n, n \geq 1$   
 $a^n b^n$   
 $a^n b^n$   
 $\Rightarrow a^+ b^+$

Which of the following grammar generates a regular language?

**A**

$G_1$  only

**B**

$G_2$  only

**C**

Both  $G_1$  and  $G_2$

**D**

None of these

$$a' = a \quad \begin{cases} 27 \\ 69 \times 9 \end{cases}$$

#Q. Consider the following three languages:

(1)  $L = \{a^{n^n} \mid n \geq 1\}$   $\{a, a^4, a^{27}, \dots\}$

(2)  $L = \{a^{m^n} \mid m = n^2, n \geq 1\}$   $a' \{a, a^{4^2}, a^{9^3}, \dots\}$

(3)  $L = \{a^{m^n} \mid n \geq 1, m > n\}$   $\{a^{2^1}, a^{3^1}, a^{4^1}, a^{5^1}, \dots\}$

Total number of regular languages is/are \_\_\_\_.

$$\{a^2, a^3, a^4, a^5, a^6, \dots\}$$

①

$$(aa)(a)^*$$

$$a^{3^2}$$

# [MCQ]



#Q. Consider the following grammar

G:

$$G = S \rightarrow AB \mid CD \Rightarrow (aa)^*b^* + (aa)^*b^* \Rightarrow (aa)^*b^*$$

$$A \rightarrow aaA \mid \epsilon \Rightarrow (aa)^*$$

$$B \rightarrow bB \mid \epsilon \Rightarrow b^*$$

$$C \rightarrow aaC \mid \epsilon \Rightarrow (aa)^*$$

$$D \rightarrow bD \mid \epsilon \Rightarrow b^*$$

ANY Lang if DFA, NFA, RE, RG

B

The language generated by above grammar is:

bD  
bbD  
bbbD

**A**

Finite ~~X~~

**B**

Infinite but regular

**C**

Non-regular ~~X~~

**D**

None of these

# [MCQ]



#Q. Which of the following language is non-regular?

**A**

$$L = \{a^{2m} b^n b^n \mid m, n \geq 1\}$$

$a^2, a^4, a^6, a^8$

$(aa)^+$

$$b^+ b^+ = (aa)^+ b b b^* = (aa)^+ b b^+$$

**B**

$$L = \{a^m b^n X \mid m, n \geq 1, X \in \{a, b\}^*\}$$

$$a^+ b^+ (a+b)^*$$

**C**

$$L = \{\{a^{n^2}\}^* \mid n \geq 0\}$$

$$\Rightarrow \{a^{1^2}\}^*, \{a^{2^2}\}^* \dots$$

**D**

None of these.

$$\Rightarrow \{a^4\}^* = \{ \epsilon, a, aa, a^3, \dots \}$$

#Q. Regular expression can be used in:

**A**

Lexical Analysis

(a.b)

**B**

Pattern matching

**C**

String matching

ag ( ) ab

**D**

Syntax analysis

## [MCQ]



#Q. Consider the regular expression:

regular expression =  $a^*b(a + ba^*)^*$

Above regular expression is equivalent to which of the following below regular expression?

**A**

$ba^*(bb)^*$

~~X~~

**B**

$ba^*(a + ba^*b)^*$

~~X~~

**C**

$(b + aa^*b) + (b + aa^*b)(ba^*b + a)(ba^*b + a)^*$

**D**

$a^*b(a + b)^*$

~~✓~~

$a^*b(a + b)^*$   
 $\{b, ab, aabb, \dots\}$



## [MCQ]



#Q. Which of the following statement will generate finite language?

**A**

~~PDA with finite stack.~~



**B**

Regular expression without kleene star and kleene plus.

$(aa + ab)$

**C**

~~Regular expression with unary alphabet.~~

$\{a\}^*$

**D**

~~Regular expression with binary alphabet.~~

$\{0,1\}^*$

## [MCQ]



#Q. Consider following regular expressions:

[I]  $(ab)^*a = a(ab)^*$

[II]  $(bb)^*b^* = b^*$

[III]  $(b + \epsilon)^+ = b^*$

Which of the following is correct?

**A**

II and III only.

**B**

I and II only.

**C**

All are correct.

**D**

None of these are correct.

## [MCQ]



#Q. For  $L = \{a^n b^m \mid n, m \geq 0\}$

What will be the regular expression ?

**A**  $(a^*b^*)^*$   $\Rightarrow (a+b)^*$

**C**  $(ab)^*$   $\Rightarrow \underline{abab} \dots$  X

**B**  $(a^*b^*)$   $\Rightarrow$  Correct

**D**  $(b^*a^*)$   $\Rightarrow \underline{bbbaa} \dots$

## [MCQ]



#Q. Consider the following regular expressions:

(I)  $(aa + aaa)^* = aa^+$

(II)  $(a^*b(a+b)^* + (a^*b^*)^*) = (a+b)^*$

(III)  $(\epsilon + aaa(aaa)^*)(\epsilon + a + aa) = (a + aa + aaa)^*$

Which the following is correct?

A

(I) and (III) only.

B

(II) and (III) only.

C

All are correct.

D

None of these are correct.

#Q. Which of the following is/are regular expression for the language:  
 $L = \{\text{Containing } \underline{ab} \text{ as a substring}\}$

$L = \{ab, .abb\}$

bab

(a, c, d)

~~A~~

$b^*aa^*b(a^*b^*)^*$

$\Rightarrow b^*aa^*b(a+b)^*$   
 $\in a \in b \in = ab$

B

$(a+b)^*(ab)^*(a+b)^*$

$\in \in \in = \in$

~~C~~

$(a^*b^*)^*ab(a^*+b^*)^*$

$(a+b)^* \underline{ab} (a+b)^*$

~~D~~

$(a+b)^*ab(a+b)^*$



# [MCQ]



#Q. What will be the regular expression for  $L = \{a^{2n} \mid n \geq 15\}$  over  $\Sigma = \{a\}$

$$\{a^{2 \cdot 15}\} = a^{30}, a^{32}, a^{34}, \dots$$

$$= a^{30} (aa)^*$$

$$a^{30} \cdot e = a^{30}$$

$$= a^{30} aa = a^{32}$$

- A**  $a^{15}(aa)^*$  X
- B**  $(aa)^* a^{15}$  X
- C**  $a^{30}(aa)^*$  ✓
- D** None of these

# [MCQ]



B

#Q. Which of the following string does not belong to  $(ab^*)^*$ ?

**A** aaabbaa

**B** baaaabb

**C** aaabbb

**D** ababa

$ab^*ab^*ab^*ab^*$   
aaabbaa

$(ab^*)^*$   
aaabbaa

$ab^*ab^*ab^*$   
ababab

14 Q

All Ques come from CN

WT, DPP, (PYQ) → 90% CN

**THANK - YOU**

Revision, PCN,